MILITARY SPECIFICATION

WARHEAD, GUIDED MISSILE, HE, M207E2
BODY AND BOATTAIL BUSHING FOR

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the drawings for the fabrication and packing of body and boattail bushing for the M207E2 Warhead.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	Ammunition Standard Quality Assurance provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-1169	Packaging, packing and Marking for Shipment of Inert Ammunition Components.

FSC: 1336

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Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9340751	Body
9333753	Bushing, Boattail

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on Drawings 9340751 and 9333753 all associated drawings and with all requirements of applicable standards and specifications.

3.3 Data requirements. The contractor shall generate data in accordance with requirements of data item description cited in 6.4.

3.4 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. All parts shall be fabricated in a thorough, workmanlike manner. The parts shall be free of burrs, chips, sharp edges, cracks, unblended radii, dents, tool marks, chipped edges, porosity, warpage, gouges, surface defects, dirt, grease, corrosion products, burn marks, and other foreign matter. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents. Surface coatings shall be continuous except for scratches not exposing base metal.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detailed specification.

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4.2 Classification of inspections. The following types of inspection shall be conducted.

a. First article inspection.

b. Quality conformance inspection.

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Body	9340751	20
Bushing, Boattail	9333753	20

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Tow Warhead M207E2 Parts				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body (Dwg. 9340751)				
	Examination for Defects	20	-	3.2	4.4.2.1
	Bushing, Boattail (Dwg. 9333753)				
	Examination for Defects	20	-	3.2	4.4.2.2

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality Conformance Inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots shall contain:

a. Body (dwg. 9340751) from not more than one interfix lot number from one manufacturer.

b. Bushing, Boattail (dwg. 9333753) from not more than one interfix lot number from one manufacturer.

4.4.2 Examinations. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.1	Body				9340751
					NEXT HIGHER ASSEMBLY 9340754
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Runout on inside surface between two basic dimensions with datum		0.40%	3.2	Gage
102	Large inside diameter (I.D.)		0.25%	3.2	Gage
103	Runout of O.D. with datums		0.40%	3.2	Gage
104	Perpendicularity of front face with datum		0.40%	3.2	Gage
105	Length to large basic dimension, aft end		0.40%	3.2	Gage
106	Diameter of small holes, front end		0.25%	3.2	Gage
107	True position of small holes, front end		100%	3.2	Gage
108	Perpendicularity of aft shoulder with O.D.		0.40%	3.2	Gage
109	Inside corner radius, large aft mounting shoulder		0.40%	3.2	Gage
110	Length between two basic dimensions, aft end		0.40%	3.2	Gage
111	Length of ring retaining full thread		0.40%	3.2	Gage
112	Pitch diameter of large aft thread		0.40%	3.2	Gage
113	Small I.D. at aft end		0.40%	3.2	Gage
114	Runout of small I.D. at aft end with datums		0.40%	3.2	Gage
115	True position of small aft holes		100%	3.2	Gage
116	Pitch diameter of small internal threads		0.40%	3.2	Gage
117	Mounting diameter, aft end		100%	3.2	Gage

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.1	Body				9340751
					NEXT HIGHER ASSEMBLY 9340754
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MAJOR (Continued)					
118	Runout of mounting diameter aft end with datum		0.40%	3.2	Gage
119	Location of conduit slot		0.25%	3.2	Gage
120	Location of boattail positioning slot		0.40%	3.2	Gage
121	True position of conduit holes		0.40%	3.2	Gage
MINOR					
201	Distance from front face to missile mounting diameter face		0.65%	3.2	Gage
202	Distance from front face to shoulder		0.65%	3.2	Gage
203	Distance from front face to inside shoulder		0.65%	3.2	Gage
204	Overall length		0.65%	3.2	Gage
205	Minor diameter of small aft threads, max.		0.65%	3.2	Gage
206	Major diameter of large aft threads, min.		0.65%	3.2	Gage
207	Wall thickness, aft end		0.65%	3.2	Gage
208	Protective finish missing or inadequate		0.65%	3.2	Visual
209	Diameter of conduit holes		0.65%	3.2	Gage
210	C' bore flush with 10° surface to .005" below		0.65%	3.2	Visual
211	Identification missing or inadequate		0.65%	3.2	Visual
212	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Boattail Bushing		9333753
			NEXT HIGHER ASSEMBLY 9340754

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Width of Warhead positioning tab		0.40%	3.2	Gage
102	Height of Warhead positioning tab		0.40%	3.2	Gage
103	Width of S&A positioning tab		0.40%	3.2	Gage
104	Height of S&A positioning tab		0.40%	3.2	Gage
105	Height of boattail bushing flange		0.40%	3.2	Gage
106	Inside diameter		0.40%	3.2	Gage
MINOR					
201	Evidence of poor workmanship		1.00%	3.5	Visual

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4.4.3 Testing. Testing is described in First Article and Quality Conformance inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1 and 4.4.2.2. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspection. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Section 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

5. PACKAGING

5.1 Packaging and packing. The parts shall be packaged and packed in accordance with the requirements of MIL-STD-1169.

5.2 Marking. Marking shall be in accordance with the requirements of MIL-STD-1169.

6. NOTES

6.1 Intended use. The body and boattail bushing covered by this specification are components of the Improved TOW Warhead Section, Guided Missile, HE, M207E2.

6.2 Ordering data. See MIL-A-48078.

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6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Data requirements. Deliverable data required by the specification is cited as follows:

Data Requirement	Applicable Data Item
Ammunition Data	DI-L-1410

Custodian:
Army-AR

Preparing Activity
Army-AR

Project Number: 1336-A401

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